Ch 6.3: Dimension Reduction

Lecture 20 - CMSE 381

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Weds, Oct 27, 2023

Announcements

Last time:

- Exam #2

This lecture:

- PCA / PCR
- ~~PLS~~

Announcements:

- Homework #6 ..... posted?

↳ Plan due Friday 11/3

Section 1

Last time

Shrinkage

Find β to minimize:

Sparse: ... fewer variables used

Least Squares:

$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

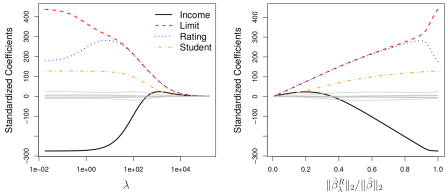
$\frac{1}{n} RSS = MSE$

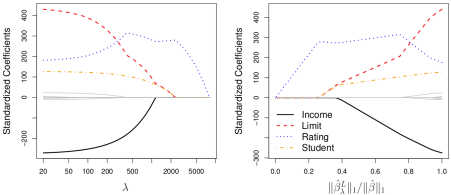
Ridge:

$$RSS + \sum_{j=1}^{p} \beta_j^2$$

The Lasso:

$$RSS + \sum_{j=1}^{p} |\beta_j|$$





Section 2

Dimension Reduction

Linear transformation of predictors

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, Z_M$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

$m = 1, \ldots, M$

Goal $p \gg M$

$$Z_m = \varphi_{1m} X_1 + \varphi_{2m} X_2 + \cdots + \varphi_{pm} X_p$$

An example or two

$p = 2 \qquad M = 1$

$$Z_1 = X_1 + 3X_2$$

$$(Z_1) = (X_1 \;\; X_2) \underbrace{\begin{pmatrix} 1 \\ 3 \end{pmatrix}}_{\varphi_2}$$

X_1	X_2
0	1
3	4
1	1
-1	0

Z_1
4
17
4
-1

$p = 3 \qquad M = 2$

$$Z_1 = \underbrace{X_1}_{\varphi_{11}} + \underbrace{0}_{\varphi_{21}} \cdot X_2 + \underbrace{0}_{\varphi_{31}} \cdot X_3$$

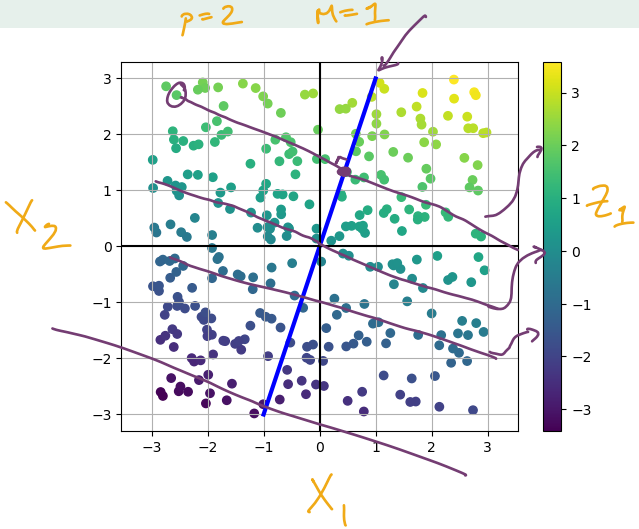
$$Z_2 = \underbrace{0}_{\varphi_{12}} \cdot X_1 + \underbrace{\tfrac{1}{2}}_{\varphi_{22}} X_2 + \underbrace{\tfrac{1}{2}}_{\varphi_{32}} \cdot X_3$$

$$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = (X_1 \;\; X_2 \;\; X_3) \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1/2 \\ 0 & 1/2 \end{pmatrix}}_{\varphi}$$

Geometric interpretation

Force φ to unit norm

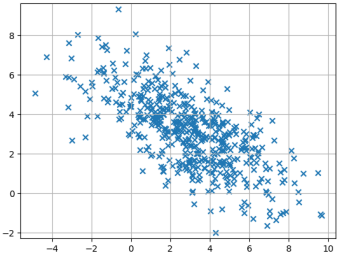
$$Z_1 = \frac{1}{\sqrt{10}} X_1 + \frac{3}{\sqrt{10}} X_2$$

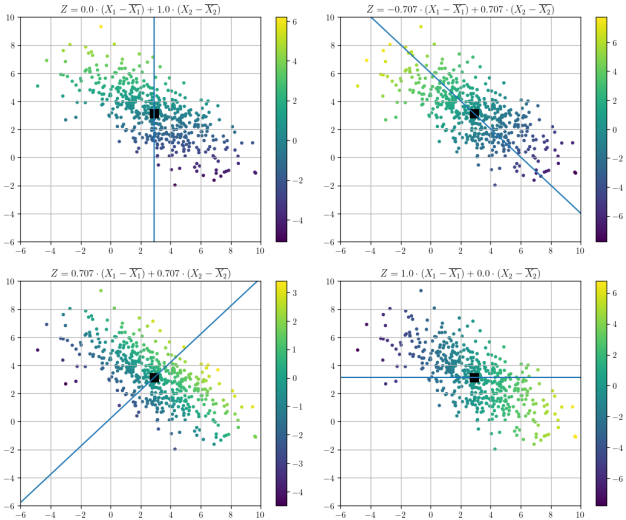


Projection onto a line

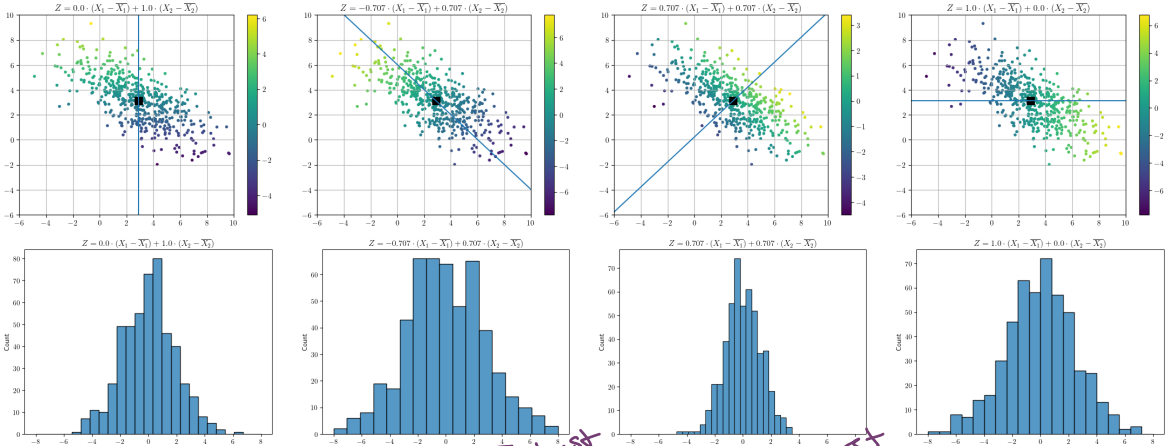
https://www.desmos.com/calculator/cih7wy8oyg

Different projections





Histograms of Z values



best

worst

Goal: Spread out the data

The goal

maximize spread of projected data

standard term

- Find good φ's for some $M \ll p$ — PCA Principal Component Analysis
- Fit regression model on Z_i's using least squares — PCR Principal Component Regression

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \varepsilon_i$$

less

Section 3

PCA

An example dataset

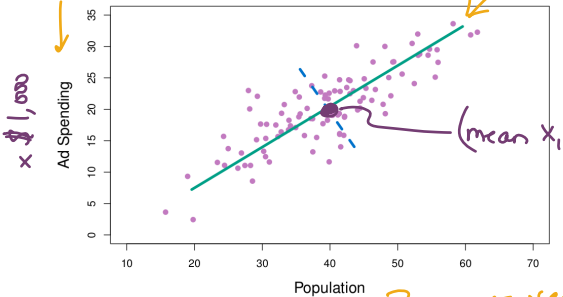
input variable

direction of max. variability

the 1st principal component

1st PC

PC1



(mean x_1, mean x_2)

× \$1,000

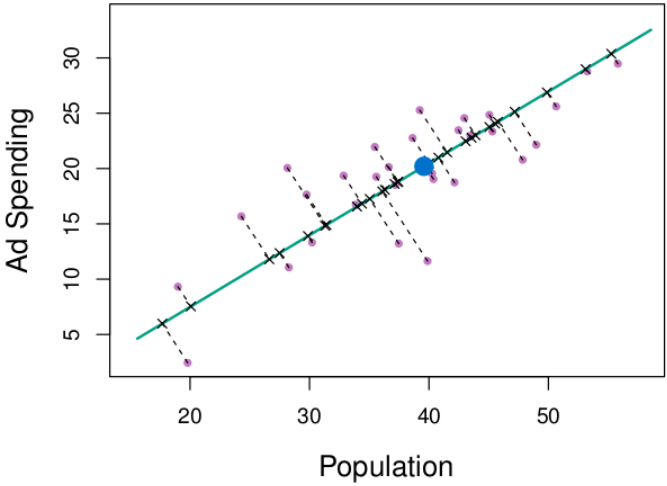
2 input variable

× 10,000 people

Projecting onto line has most variance over all choices of lines

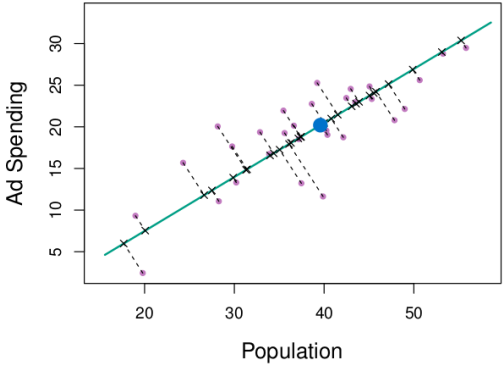
Lines have to go through $(\bar{x}_1, \bar{x}_2)$

Projection onto first PC



$$Z_1 = 0.839 \cdot (\mathrm{pop} - \overline{\mathrm{pop}}) + 0.544 \cdot (\mathrm{ad} - \overline{\mathrm{ad}})$$

What does it mean to have the highest variance



Of every combination of pop and ad this has the highest variance

$$Z_1 = \varphi_1 (pop - \overline{pop}) + \varphi_2 (ad - \overline{ad})$$

require $\varphi_1^2 + \varphi_2^2 = 1$

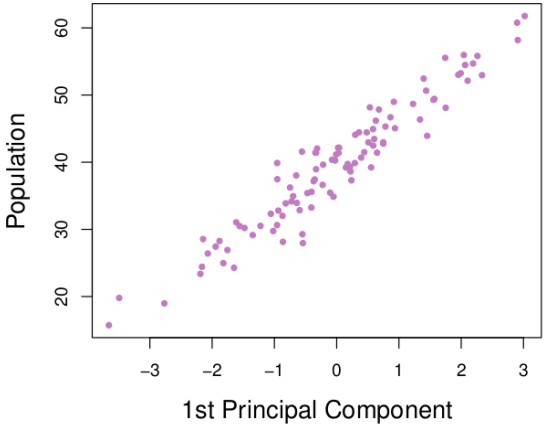
Toy for learning PCA

https://www.desmos.com/calculator/qq14tyjz0z

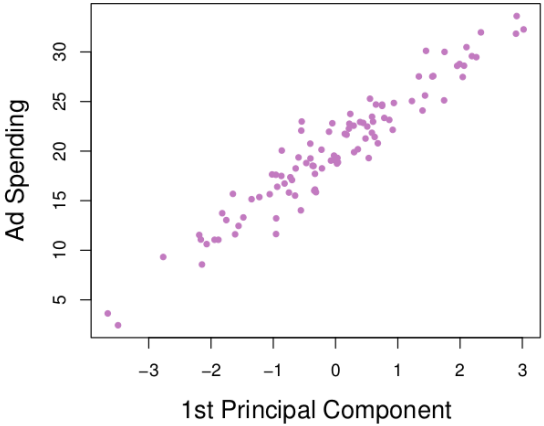
Principal component scores

ad

pop

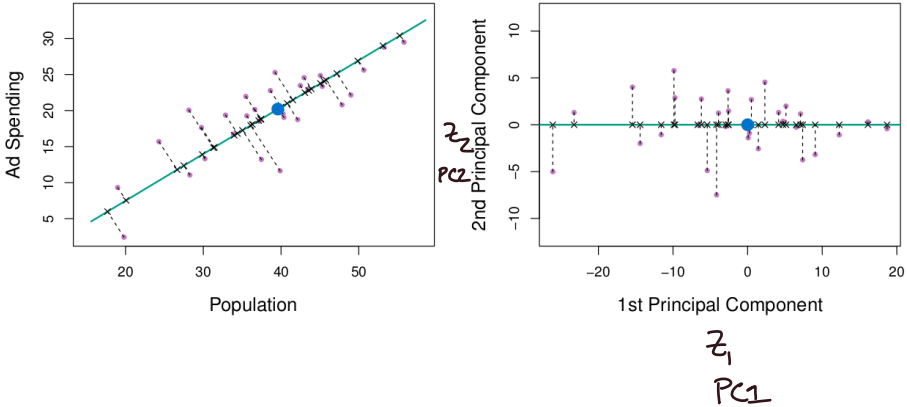


z_1



$$z_{i1} = 0.839 \cdot (\text{pop}_i - \overline{\text{pop}}) + 0.544 \cdot (\text{ad}_i - \overline{\text{ad}})$$

Another view



The other principal components

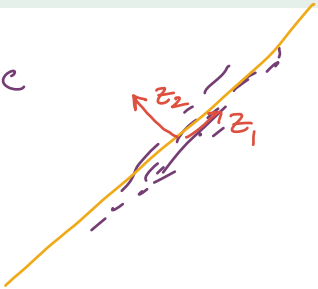
Have Z_1 1st PC

Z_2 (2nd PC) is

linear combo of

$X_1, \ldots, X_p$

uncorrelated to Z_1



3D

options for Z_2

Find one w/ most variance

$X_1, \ldots, X_p$

$\Downarrow$

$Z_1, \ldots, Z_p$

Can have p-PC's

But don't use them all!

Do PCA with Penguins

Section 4

Principal Components Regression

PCA + Linear regression

So you've found your PCA coefficients

Now what?

Have $z_1, \ldots, z_M$

$\Downarrow$

Do linear regression

$$y = \theta_0 + \theta_1 z_1 + \theta_2 z_2 + \ldots + \theta_M z_M$$

$\downarrow$

passin matrix

$$\begin{array}{c} \\ 1 \\ \vdots \\ n \end{array} \begin{array}{c} z_1, \ldots, z_M \\ \begin{pmatrix} \quad & \quad \\ \quad & \quad \end{pmatrix} \end{array}$$

to linear reg

What are we assuming?

The directions where $x_1, \ldots, x_p$ have the most variance are the ones most associated with y.

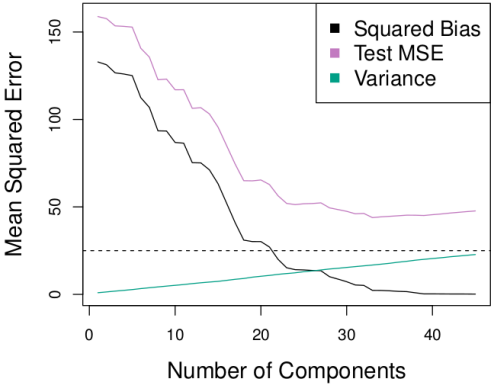
If this is true, fitting on z_i's is better than on x_i's : less overfitting!

Do PCR with hitters data

Doing better

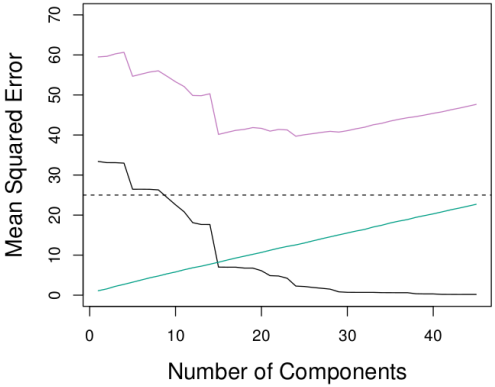
Example with simulated data: $n = 50$ observations of $p = 45$ predictors

Y is a function of 2 predictors



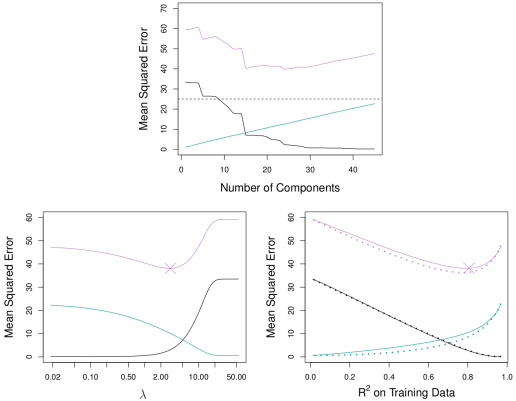
Doing better

Example with simulated data: $n = 50$ observations of $p = 45$ predictors
Y is a function of all predictors

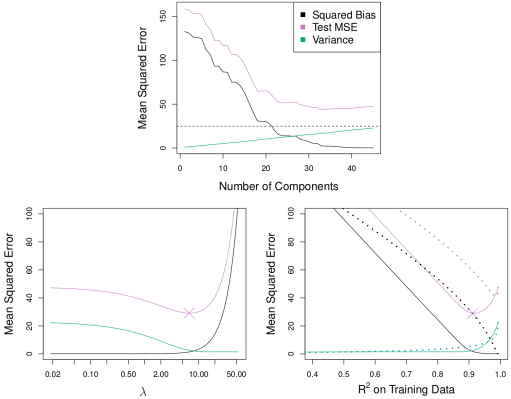


Comparison to results on shrinkage

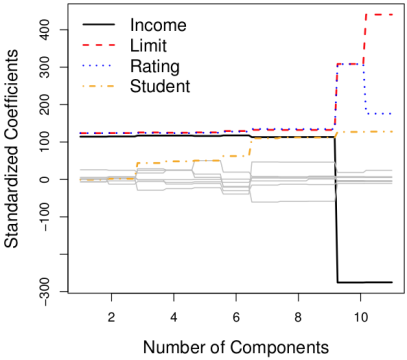
Y is a function of all predictors

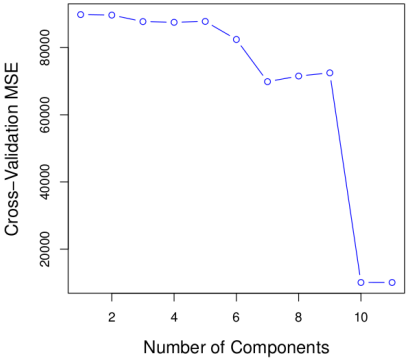


Y is a function of 2 predictors



Picking M


Income
Limit
Rating
Student
Standardized Coefficients
400
300
200
100
0
−100
−300
2
4
6
8
10
Number of Components

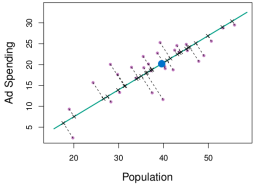

Cross−Validation MSE
80000
60000
40000
20000
2
4
6
8
10
Number of Components

Properties of PCR

TL;DR

PCA

- Unsupervised dimensionality reduction
- Choose component Z_1 in the direction of most variance using only X_i's information
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained



PCR

- Do PCA on input data
- Do Linear Regression on chosen number of PCs.
- Warning: Lose interpretability of the coefficients.

Next time

Lec #	Date			Reading	Homeworks
20	Fri	Oct 27	Dimension Reduction	6.3	
21	Mon	Oct 30	More dimension reduction; High dimensions	6.4	
22	Wed	Nov 1	Polynomial & Step Functions	7.1,7.2	
23	Fri	Nov 3	Step Functions	7.2	HW #6 Due
24	Mon	Nov 6	Basis functions, Regression Splines	7.3,7.4	
25	Wed	Nov 8	Decision Trees	8.1	
26	Fri	Nov 10	Random Forests	8.2.1, 8.2.2	
27	Mon	Nov 13	Maximal Margin Classifier	9.1	
28	Wed	Nov 15	SVC	9.2	
29	Fri	Nov 17	SVM	9.3, 9.4	
30	Mon	Nov 20	Single layer NN	10.1	
31	Wed	Nov 22	Overflow/project day?		
	Fri	Nov 24	No class - Thanksgiving		
	Mon	Nov 27	**Review**		
	Wed	Nov 29	**Midterm #3**		
32	Fri	Dec 1	Multi Layer NN	10.2	
33	Mon	Dec 4	CNN	10.3	
34	Wed	Dec 6	Unsupervised Learning & Clustering	12.1, 12.4	
35	Fri	Dec 8	Overflow/Project day?		Project due